October University for Modern Sciences and Arts (MSA)

Electronic and Communication Department

Door Security System

Design and Implementation of

4-BIT DOOR SECURITY SYSTEM WITH CHECK BUTTON

Term Project for ECE264/CSE152 Digital Logic Design Il

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A. SYSTEM SPECIFICATIONS

The system is a 4-bit door security system enhanced with a dedicated **CHECK pushbutton**. The user sets a 4-bit password using four input switches, then presses the CHECK button to trigger the comparison. Only after the button is pressed does the system evaluate the password and display the result through a **Green LED** (correct) or a **Red LED** with **Buzzer** (incorrect).

Table 1 System specifications

Password length	4 bits — 16 possible combinations (0000_2 to 1111_2)
Input switches	4 SPDT toggle switches: S3, S2, S1, S0 (user-entered password)
Preset switches	4 DIP switches: P3, P2, P1, P0 (correct password, hardwired)
CHECK button	1 pushbutton — triggers the password comparison when pressed
Correct output	Green LED turns ON — "Password is Correct"
Incorrect output	Red LED turns ON + Buzzer activates — "Password is Incorrect"
Circuit type	Purely combinational — no clock or flip-flops required
Check mechanism	XOR gate used to gate the output with the CHECK button signal
Operating voltage	5V DC regulated (via LM7805 voltage regulator)
Logic family	74HC series CMOS ICs (XOR, NOT, AND)
Response time	Gate propagation delay only (nanosecond range)
Security ratio	1 correct combination out of 16 (6.25%)

B. LOGIC DIAGRAM

The system is implemented using three logic ICs: the **74HC86** (Quad XOR), the **74HC04** (Hex NOT), and the **74HC08** (Quad AND). A fourth XOR gate from the 74HC86 is used to gate the final MATCH output with the **CHECK pushbutton** signal. The complete logic diagram is shown in Fig. 1.

How the Check Button Works

Without pressing CHECK, the system produces no output regardless of the switch positions. When CHECK is pressed, its signal (logic HIGH) is combined with the **MATCH signal** using an additional XOR gate configured as a controlled buffer. The outputs — **Green LED**, **Red LED**, and **Buzzer** — are only activated at the moment the button is pressed.

Boolean Expressions

The bit-wise equality checks are performed by XNOR gates (XOR followed by NOT):

$$E3 = \overline{(S3 \oplus P3)} \quad (1)$$

$$E2 = \overline{(S2 \oplus P2)} \quad (2)$$

$$E1 = \overline{(S1 \oplus P1)} \quad (3)$$

$$E0 = \overline{(S0 \oplus P0)} \quad (4)$$

The four equality outputs are combined by cascaded AND gates:

$$MATCH = E3 \cdot E2 \cdot E1 \cdot E0 \quad (5)$$

The CHECK button gates the output. Let CHECK = pushbutton signal (HIGH when pressed):

$$ENABLED = MATCH \cdot CHECK \quad (6)$$

The final outputs (in this case) are:

GREEN LED = ENABLED (HIGH when MATCH=1 and CHECK pressed) (7)

RED LED + BUZZER = ENABLED’ = NOT(ENABLED) (8)

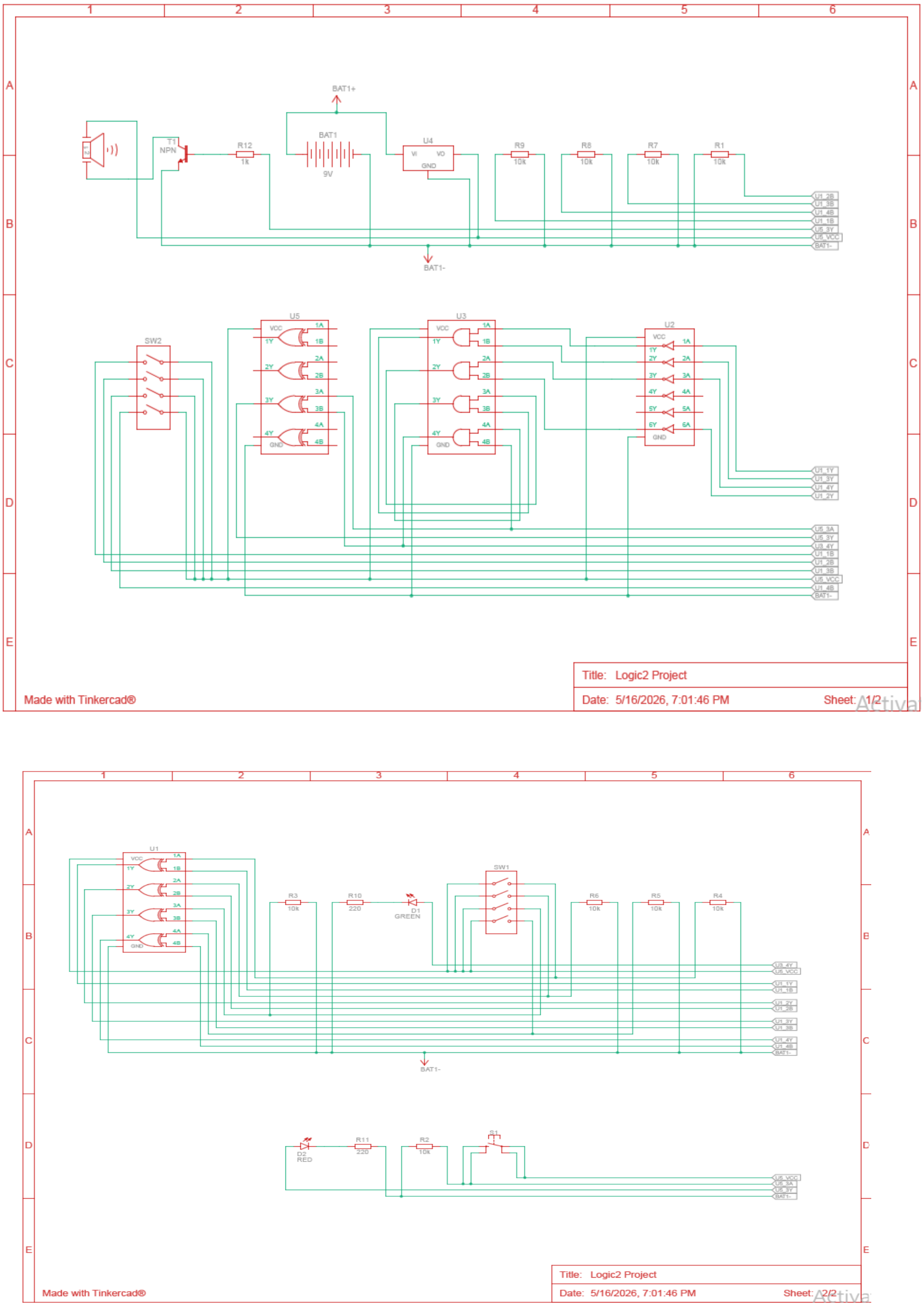


Fig. 1. Logic Diagram with ICs, Switches, Leds, Buzzer, Transistor.

Table 2 IC functions summary

IC	Part Number	Function in This Circuit	Gates Used
IC1	74HC86	XOR gates for bit comparison + CHECK button gate	4 of 4
IC2	74HC04	NOT gates: invert XOR → XNOR; invert ENABLED → Red LED/Buzzer	5 of 6
IC3	74HC08	Gates 1–2: pair AND; Gate 3: final MATCH; **Gate 4: MATCH · CHECK = ENABLED**	3 of 4
IC4	74HC86	XOR gate for inverting the (ENABLED) zero case to turn on the Red Led and Buzzer	1 of 4

C. OUTPUT SIMULATION RESULTS

The circuit was simulated in Tinkercad. The preset password was set to 1010_2 (decimal 10). Three simulation scenarios were tested: button not pressed, incorrect password with button pressed, and correct password with button pressed. Results are summarised in Table 3.

Table 3 Simulation results (preset = 1010_2)

Scenario	Input (S3–S0)	CHECK Pressed?	MATCH	Green LED	Red LED + Buzzer
No button press	1010	No	1	OFF	OFF
Wrong password + CHECK	0111	Yes	0	OFF	ON
Wrong password + CHECK	1111	Yes	0	OFF	ON
Correct password + CHECK	**1010**	**Yes**	**1**	**ON**	**OFF**
Wrong (1 bit off) + CHECK	1011	Yes	0	OFF	ON

- The simulation confirms that pressing CHECK is required to activate any output. Only the exact password 1010_2 with CHECK pressed illuminates the **Green LED**. All other combinations with CHECK pressed activate the **Red LED** and **Buzzer**.

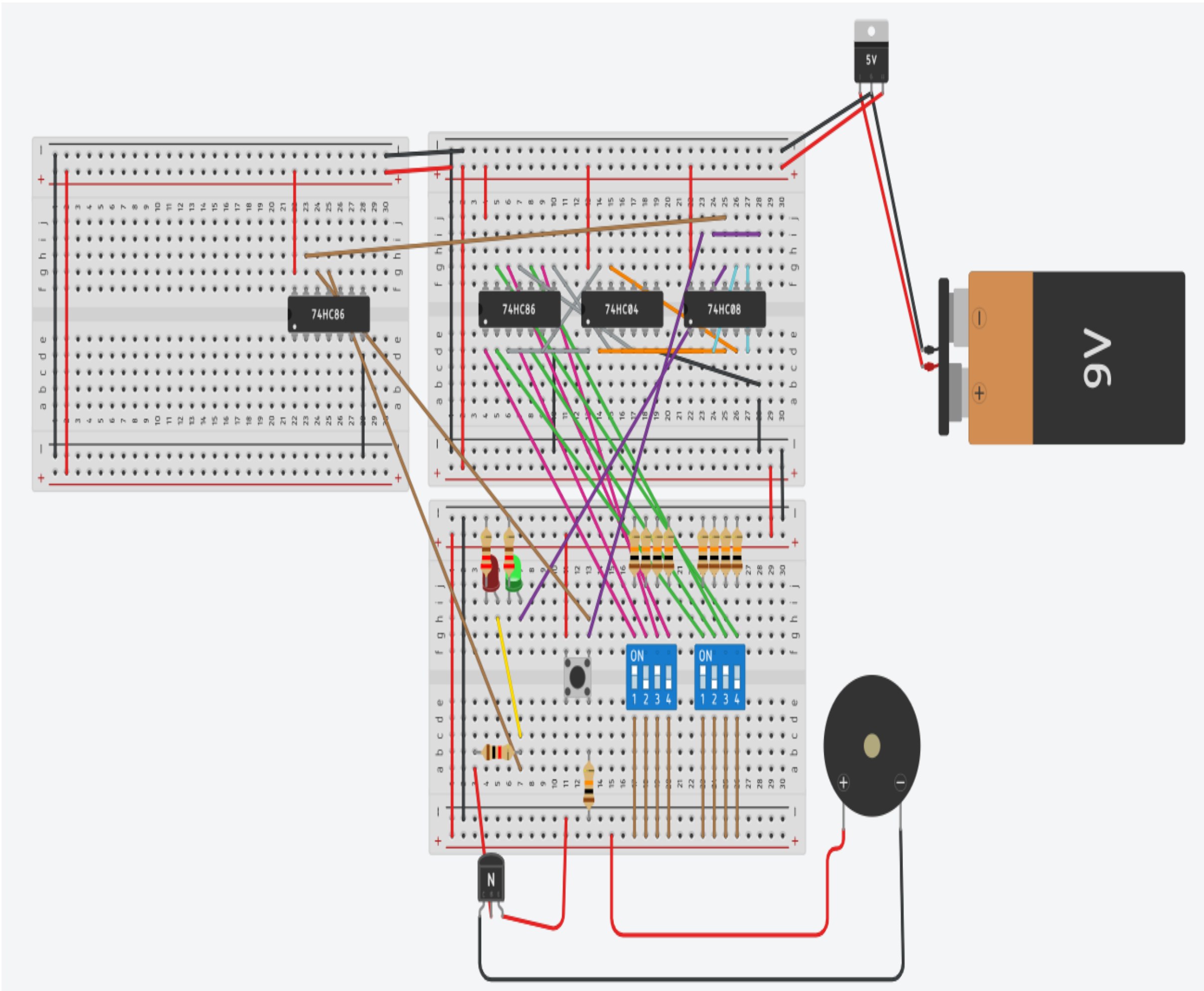
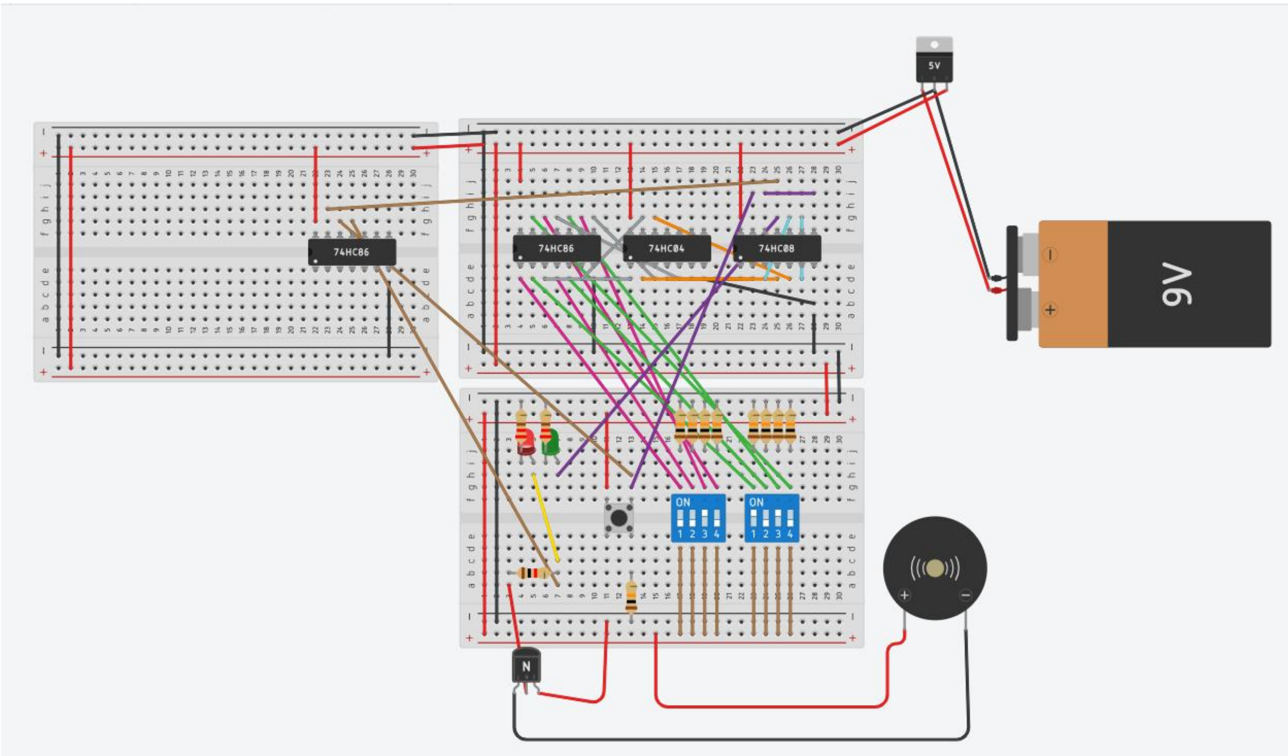


Fig. 2. Simulation result — correct password entered, CHECK pressed (Green LED ON).

Fig. 3. Simulation result — incorrect password entered, CHECK pressed



(Red LED and Buzzer ON)

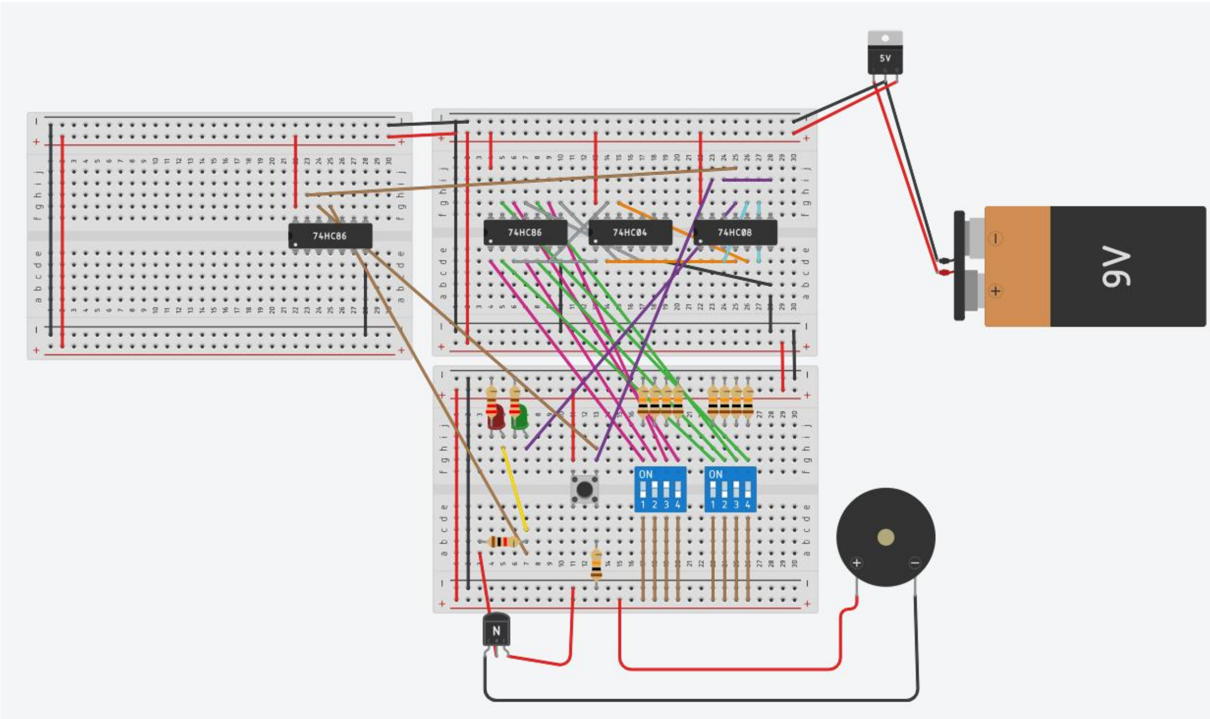


Fig. 4. Simulation result — CHECK button not pressed, all outputs remain OFF

D. LAB TEST RESULTS

The hardware circuit was assembled on a breadboard and powered by a 5V regulated supply using the LM7805 voltage regulator. The preset password was set to 1010_2. The following test cases were applied, and the outputs were recorded as shown in Table 4.

Table 4 Lab test results (preset = 1010_2)

Test	Input (S3–S0)	CHECK	Expected Output	Actual Output	Pass/Fail
1	1 0 1 0	Pressed	Green LED ON, Red LED OFF, Buzzer OFF	[Green LED on]	PASS
2	0 1 1 1	Pressed	Red LED ON, Buzzer ON, Green OFF	[Red LED, Buzzer are off]	PASS
3	0 0 0 0	Pressed	Red LED ON, Buzzer ON, Green OFF	[Red LED, Buzzer are off]	PASS
4	1 1 1 1	Pressed	Red LED ON, Buzzer ON, Green OFF	[Red LED, Buzzer are off]	PASS
5	1 0 1 0	Not pressed	All outputs OFF	[No Change]	PASS

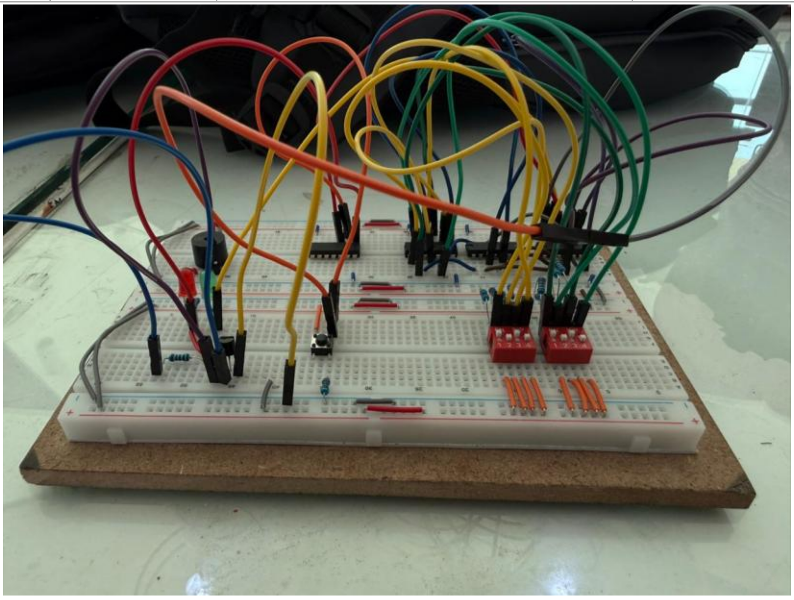


Fig. 5. Breadboard hardware implementation

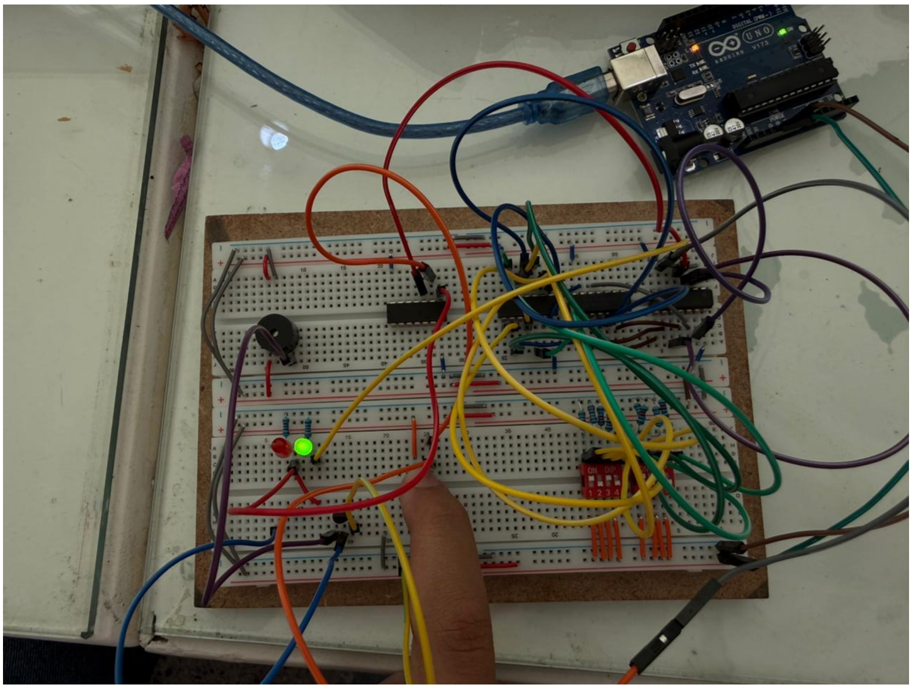


Fig. 6. Lab test — correct password entered, CHECK pressed (Green LED ON)

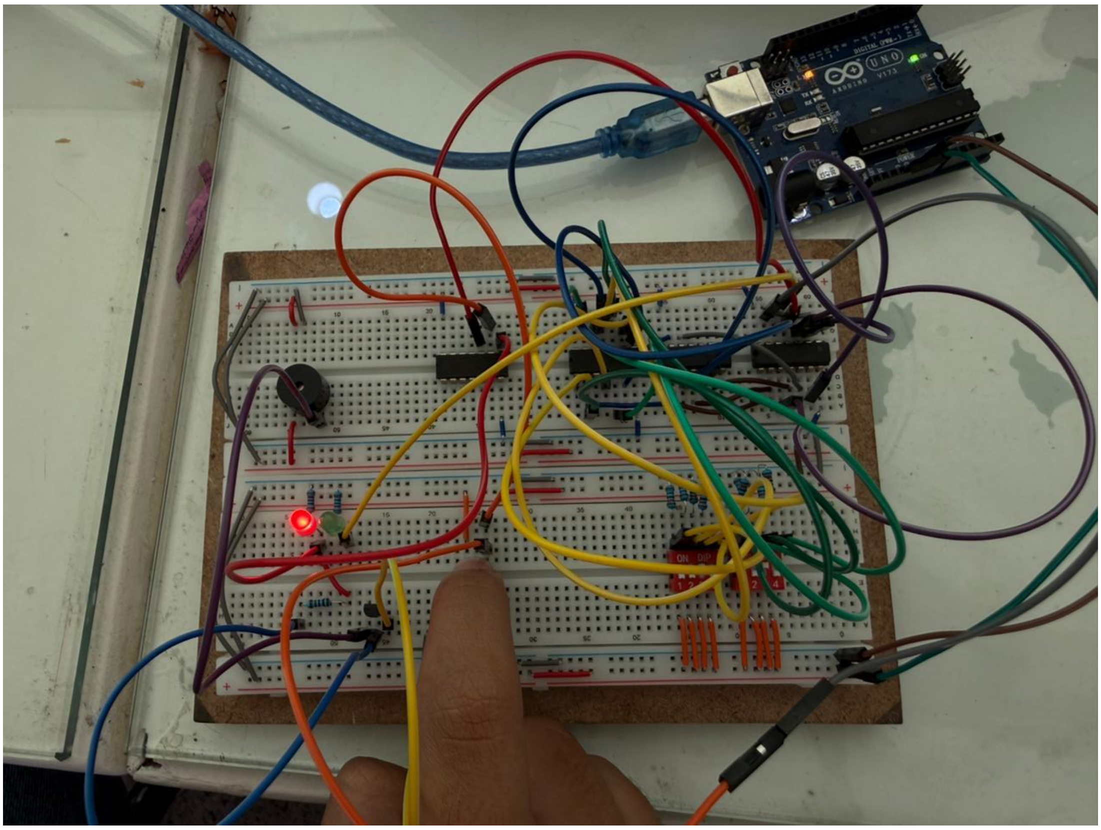


Fig. 7. Lab test — incorrect password entered, CHECK pressed (Red LED and Buzzer ON)

REFERENCES

[1] Thomas L. Floyd, “Digital Fundamentals,” 10th Edition, Prentice-Hall, 2014.

[2] M. Morris Mano, “Digital Logic and Computer Design,” 4th Edition, Pearson.

[3] Texas Instruments, “SN74HC86 Quadruple 2-Input Exclusive-OR Gate Datasheet,” Texas Instruments Inc., 2023.

[4] Texas Instruments, “SN74HC04 Hex Inverters Datasheet,” Texas Instruments Inc., 2023.

[5] Course Lecture Notes — Digital Logic II, MSA University, Faculty of Engineering, 2025–2026.